OpenDC-LCA: an evidence-governed framework for transferable life-cycle assessment of data-center cooling

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Abstract

Liquid cooling can reduce data-center operating energy and water use, yet the preferred architecture depends on electricity supply, compute performance, hardware and fluid inventories, service life, climate, and the functional unit used for comparison. Existing studies establish the importance of these effects but are difficult to update or transfer because foreground bills of materials and background processes are often proprietary, operational performance is represented by annual averages, water consumption is rarely scarcity-weighted, and data-quality assessments are not connected to measurement priorities. We present OpenDC-LCA, an open-source, evidence-governed research layer that couples data-center cooling physics to general life-cycle assessment (LCA) tools while preserving source identity, functional unit, geography, transformation equations, uncertainty, and evidence status. A guided practitioner workflow converts a compact facility, cooling, electricity, water, equipment, and source record into the governed scenario schema and returns both a plain-language report and machine-readable results with contribution breakdowns, audit findings, evidence level, and next-data requirements. The framework supports openLCA JSON-LD and Brightway interoperability, hourly weather-load-performance integration, discrete and reliability-driven replacement, uncertainty analysis, and a machine-readable claim gate. We use released Microsoft/WSP results, EPA eGRID 2023, 48 Boavizta server product-carbon-footprint records, selected ÖKOBAUDAT construction processes, NOAA weather data, and the released Microsoft pedigree assessment to demonstrate analyses that cannot be obtained from any source alone. All 24 released normalized totals were independently reconciled to floating-point precision. Re-basing the released greenhouse-gas contributions across 51 eGRID state records identified a cold-plate/one-phase-immersion crossover near 61 kg CO_2e MWh^{-1}: cold plate was lower only for Vermont among the observed states, whereas two-phase immersion remained lowest under the released model. Across the state range, embodied contributions increased from approximately 4% to 57-58% as electricity intensity declined. A contribution-weighted pedigree diagnostic ranked use-phase performance first and server inventories next as evidence-improvement priorities. Public server records showed a manufacturing median of 1,215 kg CO_2e per server and a 465-2,503 kg CO_2e range. ÖKOBAUDAT unit-process comparisons indicated 60% lower A1-A3 GHG for high-scrap electric-arc-furnace steel than a blast-furnace route and 50% lower GHG for CEM III than CEM II/A cement, but facility propagation is blocked until quantities are disclosed. The results show that open data can support new, decision-relevant deductions when incompatible units and evidence limits are not hidden. OpenDC-LCA complements rather than replaces openLCA or Brightway

by supplying the cooling-specific engineering, evidence lineage, and claim governance needed for reproducible comparative research.

Keywords: data center; cooling; life-cycle assessment; PUE; immersion cooling; liquid cooling; reproducibility; open-source software

1. Introduction and literature review

1.1. Cooling is becoming a lifecycle design decision

Data-center electricity demand is increasing with cloud and artificial- intelligence workloads even as energy per computation improves [14,15]. Simultaneously, processor heat flux and rack power density are moving beyond the practical envelope of conventional air cooling. Cold plates transfer heat from selected high-power devices to a liquid loop; single-phase immersion submerges the server in a dielectric liquid; and two-phase immersion uses boiling and condensation. Reviews document substantial differences in heat- transfer coefficient, allowable chip temperature, parasitic power, heat- rejection design, water use, maintainability and deployment maturity [16,17]. Experimental work further shows that cooling can affect clock frequency and compute density, meaning that equal IT electricity or equal server count may not represent equal service [18,19].

These interactions make a cooling comparison fundamentally different from a component efficiency comparison. A design that lowers fan energy can require tanks, cold plates, coolant distribution units, pumps, piping or fluorinated fluid. A design that permits overclocking may deliver more virtual cores but alter power, server count and lifetime. A design that eliminates evaporative cooling can reduce onsite water while changing electricity-mediated water. Consequently, the comparison must connect thermal performance, facility operation, equipment and fluid production, replacement, end of life, grid conditions and a functionally equivalent computation unit.

1.2. What existing data-center LCAs establish

Early screening LCA work showed that operational metrics such as power usage effectiveness (PUE) are insufficient because interventions can transfer burdens between operation, construction and equipment production [20]. Facility-level studies subsequently quantified the importance of electricity, servers, buildings and cooling systems. Siddik, Shehabi and Marston mapped the energy, carbon and water footprint of U.S. data centers and demonstrated that location changes the trade-off between energy- and water-related objectives [21]. Ristic et al. separated onsite water from electricity-supply-chain water and emphasized the uncertainty of water factors [22]. Product-focused work used manufacturer inventories to assess the manufacturing of a cooling device across multiple environmental categories [23], while Wenzel et al. compared LCA with material-flow, exergy and life-cycle-exergy approaches for a data-center cooling tower [24].

Alissa et al. provided the most comprehensive public comparison of air, cold-plate, single-phase immersion and two-phase immersion cooling at hyperscale [1]. Their cradle-to-grave model included buildings, supporting equipment, servers, racks or tanks, cables, fluids, use-phase electricity and water, replacements and end of life. Crucially, they used Vcore-year rather than facility area or IT energy, allowing cooling-enabled overclocking and packing density to enter the functional unit. The released results showed 15-21% GHG, 15-20% primary-energy and 31-52% blue-water reductions relative to air cooling, depending on architecture and electricity case. The study also released normalized component results, detailed equations and a pedigree assessment, creating an unusually strong basis for independent secondary analysis [2].

Recent work reinforces two conclusions. First, grid decarbonization and cooling efficiency are complementary rather than interchangeable; improving one can shift burdens or expose categories that the other does not address [25]. Second, as operational electricity becomes cleaner, manufacturing and replacement of servers and electronics become proportionally more important. ICT LCAs report large embodied impacts and methodological dispersion among products [26,27]. These results make data quality—not only model completeness—a design concern.

1.3. Water, time and geography remain weakly connected

Blue-water consumption is a volume; water-scarcity impact depends on where and when that volume is consumed. The AWARE consensus method characterizes remaining water availability and its uncertainty [28,29]. Yet data-center studies often combine onsite evaporation and power-sector water at annual, national resolution, or use watershed boundaries without scarcity factors. Likewise, annual-average PUE hides part-load, weather, humidity, economizer availability and heat-rejection behavior. Hourly cooling studies show that energy and onsite/source-water rankings can change by season and climate [30,31]. A transferable cooling LCA therefore requires both a temporal performance model and a geographically matched impact model; neither can be substituted by a site name alone.

1.4. General LCA software does not close the domain gap

openLCA and Brightway provide transparent process-network calculation, inventory exchange and life-cycle impact assessment [3,32]. Commercial tools and databases provide additional curated processes. These engines are necessary, but they do not determine whether two cooling systems deliver equivalent computation, whether a measured performance surface covers all operating hours, how reliability changes replacement burden, or whether synthetic values may support a public comparative claim. Nor does importing a dataset guarantee compatible geography, reference flow, allocation, system model or impact method.

The literature therefore leaves six connected bottlenecks:

1. released totals are not readily updateable when foreground quantities,

background mappings or licensed processes are unavailable;

2. annual-average results obscure grid, climate and part-load transferability;
3. Vcore-year, IT MWh, server-year, rack and kilogram-of-material units are

frequently discussed together without a formal conversion;

4. old or proxy electronics, support-equipment and fluid inventories become

more influential under grid decarbonization;

5. pedigree matrices describe weakness but do not identify which weak data

most affect the decision; and

6. general LCA tools lack a cooling-specific evidence and comparative-claim

contract.

1.5. Objectives and contribution

This work addresses those bottlenecks with an open, evidence-governed domain layer rather than another background database or general LCA engine. The objectives are to:

7. preserve source identifiers, functional units, transformations and evidence

status across heterogeneous cooling evidence;

8. connect static lifecycle models to measured load-weather performance,

replacement and general-purpose LCA inventories;

9. independently reconstruct the released Microsoft/WSP results and use them

with public data to derive location-dependent rankings and burden shifts;

10. combine contribution magnitude with released pedigree scores to prioritize

the next experiments and inventories; and

11. prevent numerical outputs from being communicated beyond the evidence that

supports them.

2. Framework and methods

2.1. Software and evidence architecture

Figure 1 summarizes the OpenDC-LCA workflow. Evidence may originate from published LCA results, public inventories, grid and climate datasets, laboratory measurements, or simulations. Each source record preserves its provider, persistent identifier, version, geography, reference year, declared unit, system boundary, license or redistribution constraint, evidence status, and local checksum. Harmonization is performed before calculation rather than after results are generated. Scenario JSON and performance CSV files remain human-readable and can be processed through the command line, a stable Python API, or a local graphical interface. All interfaces call the same validated calculation engine.

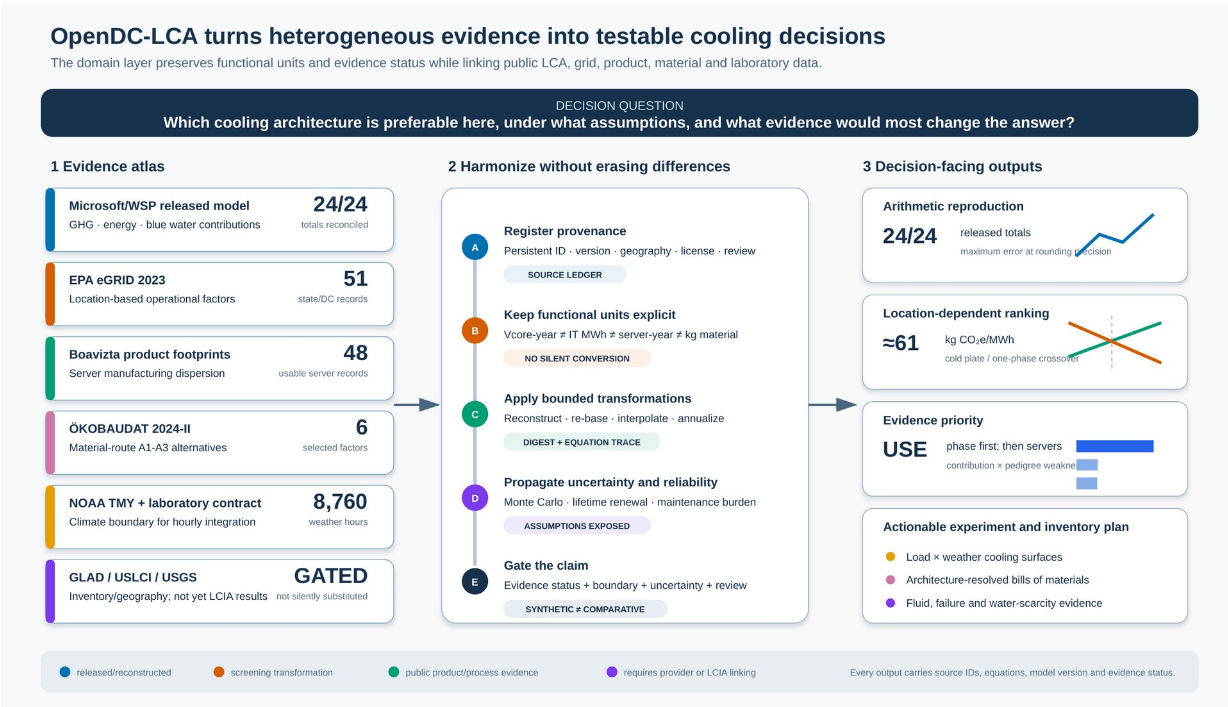

Figure 1. OpenDC-LCA evidence-to-decision architecture. The evidence atlas identifies the numerical role and limits of each downloaded source; the harmonization spine preserves provenance, functional unit, bounded transformations, uncertainty and evidence status; decision-facing outputs expose arithmetic reconstruction, geographic crossover conditions, data-improvement priorities and the next experimental or inventory need.

The framework is implemented in Python 3.10 or later and distributed under the MIT License [4]. Version 1.1.0 exposes validated scenario analysis, performance-map summarization, scientific audit functions, report generation, a local browser-based GUI, complete openLCA JSON-LD process exchange, and Brightway database-write mappings. For first-time users, a compact input contract requests IT capacity, utilization, PUE, facility lifetime, onsite water, three electricity factors, optional aggregate cooling equipment, and a citable source record. The software expands this record into the governed scenario schema before calculation; it does not maintain a separate simplified model. The resulting practitioner report includes annual and per-IT-MWh impacts, contributions, automated evidence classification, warnings or blockers, and the specific evidence required to strengthen the study. The GUI binds to the local host by default and does not alter the data model or calculation path.

2.2. Goal, functional unit, and system boundary

The present functional unit is one MWh of electricity delivered to IT equipment (`it_mwh`). This denominator is valid only when the alternatives provide equivalent computational service, utilization, reliability, and hardware life. If cooling changes compute performance, throttling, server configuration, or replacement, an IT-energy denominator alone is insufficient and absolute results plus a service-based unit should be reported.

The default `cooling_system_cradle_to_grave` boundary includes cooling and heat-rejection equipment, fluids, represented maintenance and replacement, operation, and end of life. A `facility_cradle_to_grave` option can include building, electrical, and IT assets. Common systems may be excluded only when their quantities, performance, lifetime, and end-of-life treatment are unchanged between alternatives. Location-based and market-based electricity results are kept separate. On-site water consumption and electricity supply-chain water are also reported separately.

2.3. Static energy and lifecycle calculations

For IT capacity P_IT, capacity factor u, and study duration of one year, annual IT electricity is

$$E\_IT = P\_IT\ u\ (8760) \qquad (1)$$

Annual facility electricity is

$$E\_facility = E\_IT\ PUE \qquad (2)$$

For an operational impact factor EF_k for category k, the operational impact normalized by delivered IT electricity is

$$I\_k,op = EF\_k\ PUE \qquad (3)$$

Equipment production and end-of-life burdens may be annualized with a discrete replacement model:

$$I\_k,eq = \Sigma\_i\ q\_i\ ceil(L\_s/L\_i)\ (I\_k,i\hat{}prod + I\_k,i\hat{}EOL) / L\_s \qquad (4)$$

where q_i is component quantity, L_s is the study period, and L_i is component service life. A linearized option is retained for screening. Negative end-of-life factors are permitted only for explicitly documented recovery or substitution credits.

For a cooling fluid with initial mass m_0, facility life L_f, annual loss m_loss, production factor I_k^fluid, and direct GWP GWP_direct,

$$m\_prod,annual = m\_0 / L\_f + m\_loss \qquad (5)$$

$$I\_k,fluid = m\_prod,annual\ I\_k\hat{}fluid; \quad GHG\_direct = m\_loss\ GWP\_direct \quad (6)$$

2.4. Multi-source evidence synthesis

The Microsoft/WSP normalized GHG workbook reports use-phase and embodied contributions for a U.S. grid case and a 100% renewable case. To test geographic transferability without asserting access to the proprietary background model, we used an affine re-basing. For technology j, eGRID state s, and the generation-weighted national eGRID factor EF_US,

$$I\_j,s = I\_j,RE + (I\_j,grid - I\_j,RE)\ EF\_s / EF\_US \qquad (7)$$

The published grid and renewable endpoints are therefore reproduced at ratios one and zero, respectively. This is a screening transformation, not a claim that all upstream electricity impacts scale with direct eGRID CO2e. State records with missing or nonpositive generation were excluded. The crossover between technologies a and b follows by equating the two affine relationships:

$$EF^*\_(a,b) = EF\_US\ (I\_b,RE - I\_a,RE) / [(I\_a,grid-I\_a,RE) - (I\_b,grid-I\_b,RE)] \quad (8)$$

To translate the released pedigree matrix into a research priority, each component's mean grid-case GHG contribution share \bar{s_c was multiplied by its normalized mean pedigree weakness. With pedigree score q=1 best and q=5 worst,

$$P\_c = S\_c\ [(D\_c - 1) / 4] \qquad (9)$$

P_c is a transparent screening diagnostic, not a formal expected value of information. It ranks data that are simultaneously consequential in the released model and weak according to the released assessment.

Boavizta records were filtered to `Datacenter/Server` products with total GHG, manufacturing share and lifetime. Manufacturing GHG was calculated as total product GHG multiplied by the reported manufacturing share and was not mixed with the Microsoft Vcore-year results. ÖKOBAUDAT A1-A3 processes were compared within matched product families using 100(1-I_alt/I_base). These per-kilogram reductions were not propagated to a facility because the necessary architecture-specific material quantities are not public.

2.5. Performance-map and temporal integration

Each performance-map row records architecture, test identifier, IT load, heat removed, coolant temperatures, flow, pressure drop, pump power, fan power, CDU power, heat-rejection power, on-site water rate, dry- and wet-bulb temperature, duration, measurement uncertainty, and source identifier. Aggregate metrics are energy-weighted rather than arithmetic averages of ratios.

The measurement-ready interface uses a complete rectangular surface over IT load P and ambient dry-bulb temperature T. For a query point inside one cell of the measured grid, cooling power is evaluated by bilinear interpolation:

$$Q_c(P,T) = (1-\alpha)(1-\beta)Q_{11} + \alpha(1-\beta)Q_{21} + (1-\alpha)\beta Q_{12} + \alpha\beta Q_{22} \qquad (10)$$

where α=(P-P_1)/(P_2-P_1) and β=(T-T_1)/(T_2-T_1). The same operation is applied to on-site water rate and declared measurement uncertainty. Queries outside the measured domain are rejected; the software does not silently extrapolate.

For hour h, partial PUE and normalized operational GHG are

$$PUE_h = 1 + Q_{c,h} / P_{IT,h} \qquad (11)$$

$$GHG_{op} = [\Sigma_h E_{IT,h}\, PUE_h\, EF_{grid,h}] / [\Sigma_h E_{IT,h}] \qquad (12)$$

The current public-data demonstration uses an annual location-based EPA eGRID factor rather than an hourly or marginal factor. Weather observations are from a NOAA typical meteorological year (TMY) file [5,6].

2.6. Uncertainty, sensitivity, and scientific audit

OpenDC-LCA supports seeded Monte Carlo screening with uniform, triangular, normal, and lognormal distributions for selected continuous parameters. Independent sampling improves reproducibility but does not represent correlation, systematic measurement bias, or model-form uncertainty. One-at-a-time elasticity identifies influential inputs but should not be interpreted as a global sensitivity analysis.

The audit operates on scientific as well as numerical requirements. It checks source completeness, duplicate identifiers, physical values, functional unit, boundary description, evidence status, uncertainty disclosure, and comparative assertion intent. Synthetic or illustrative data must set the comparative assertion to false. A public comparative claim additionally requires decision-grade provenance, consistent boundaries, quantified uncertainty, and independent critical review consistent with the intent of ISO 14040 and ISO 14044 [7,8]. The automated audit supports review but is not an ISO conformity assessment.

2.7. Inventory interoperability and reliability

OpenDC-LCA reads complete openLCA JSON-LD process exchanges, including flow and provider identifiers, flow type, amount, unit, direction, and quantitative reference. The same records map to dictionaries accepted by `bw2data.Database.write`. Conversely, an OpenDC-LCA scenario can be exported as an annual foreground unit process containing delivered IT service, facility electricity, on-site water, and equipment exchanges. LCIA factors are not misrepresented as elementary flows; background providers, elementary-flow mapping, and impact methods remain explicit tasks in openLCA or Brightway.

For a component with Weibull characteristic life \eta and shape β, the first-failure distribution is

$$F(t) = 1 - exp[-(t / eta)^beta] \quad (13)$$

Expected failures after as-good-as-new replacement are obtained from the renewal equation

$$M(t) = F(t) + integral[0,t] M(t-x) dF(x) \quad (14)$$

An optional Arrhenius acceleration adjusts characteristic life from reference temperature T_ref to operating temperature T_op:

$$eta\_op = eta\_ref / exp[Ea/kB (1/T\_ref - 1/T\_op)] \quad (15)$$

The calculation reports expected failures and installations, scheduled maintenance, maintenance impacts, downtime, and unserved-IT-energy exposure. These are expectation values. Redundancy, common-cause failure, repair queues, and backup-system energy are outside the present model.

3. Data and evaluation cases

3.1. Microsoft/Nature released source data

The validation case used the source data and model archive released with Alissa et al. [1,2]. The tested claim was limited: for each architecture, electricity scenario, and impact category, did the released normalized total equal the sum of the seven released contributions for use phase, building, server, rack or tank, cable, supporting equipment, and cooling fluid? The audit covered four architectures, two electricity scenarios, and three metrics, for 24 totals. It did not independently recreate licensed background inventories, manufacturer data, confidential bills of materials, measured PUE values, or the complete virtual-core model.

3.2. EPA eGRID geographic transfer case

The metric eGRID 2023 workbook supplied state net generation (`STNGENAN`) and state annual CO_2-equivalent total-output emission rate (`STC2ERTA`) [5]. The analysis included 50 states and the District of Columbia. The generation-weighted mean was 348.194 kg CO_2e MWh^{-1} and the observed state range was 23.696-893.079 kg CO_2e MWh^{-1}. These are annual, location-based factors; they are not marginal, hourly or market-based. The factors were used only in the affine re-basing of Eq. (7).

3.3. Boavizta server product-footprint case

The downloaded U.S. Boavizta table contained 1,226 ICT product records. We retained 48 `Datacenter/Server` records having total GHG, manufacturing share and declared lifetime [11]. Each record retains manufacturer, product, report date and source URL. Product carbon footprints were developed by different manufacturers and may differ in boundary, electricity, allocation and use assumptions. The distribution therefore measures public-factor dispersion; it is not a like-for-like server ranking.

3.4. ÖKOBAUDAT construction-process case

Six selected ÖKOBAUDAT 2024-II generic A1-A3 records retained UUID, product name, geography, reference year, unit and URL [10]. Matched comparisons were made between galvanized steel profiles using high-scrap electric-arc-furnace and low-scrap blast-furnace routes, and between CEM III and CEM II/A cement. Ready-mix concrete records were registered but not compared across mass units because their reference unit is cubic metre. No construction factor was added to a data-center total without a foreground quantity.

3.5. Released pedigree assessment

The Microsoft/WSP workbook contains five scores for each of nine component groups: reliability, completeness, temporal correlation, geographical correlation and technological correlation. Scores range from 1 (best) to 5 (worst). Forty-five component-criterion records were parsed with their full released rationales and combined with mean grid-case GHG contribution shares using Eq. (9).

3.6. Climate and measurement-surface software cases

NOAA TMY data for the station nearest Fayetteville supplied 8,760 hourly weather observations [6]. The public preview combined these observations with explicit piecewise-linear temperature-PUE hypotheses. Complete air-cooled and direct-to-chip load-temperature surfaces were also generated as synthetic fixtures to test input validation, bilinear interpolation, uncertainty bounds, hourly workload-weather alignment and annual aggregation. These curves and surfaces are not measurements.

Both datasets carry `evidence_status = synthetic`, and the machine-readable output sets `comparative_claim_allowed = false`. Their difference is a software test, not a finding about cooling technologies.

3.7. Registered but not numerically combined evidence

The source registry also documents USLCI v1.2026-06.0, ÖKOBAUDAT 2024-II, seven NREL hydrogen datasets accessed through GLAD and the Federal LCA Commons, and USGS watershed data [9,12,13]. These sources establish potential inputs for equipment, backup power and water context. The seven GLAD archives can now be read as complete openLCA JSON-LD inventories and mapped to Brightway; the `.zolca` USLCI backup must first be exported from openLCA as portable JSON-LD. Registration or successful exchange does not make datasets automatically compatible; geography, reference year, unit, product system, impact method, and allocation must be harmonized for each study. No numerical LCIA result was calculated from GLAD or USLCI because providers and characterization methods have not yet been linked. The USGS HU12 archive supplies watershed boundaries, not water consumption or AWARE factors, and therefore cannot support a scarcity result.

4. Results

4.1. Exact reconstruction of released normalized totals

All 24 Microsoft/Nature released totals were reconstructed by summing the seven released contribution categories. The maximum absolute disagreement was 1.42×10^{-14} percentage points, consistent with floating-point representation. Figure 2 summarizes the grid-scenario reductions relative to air cooling.

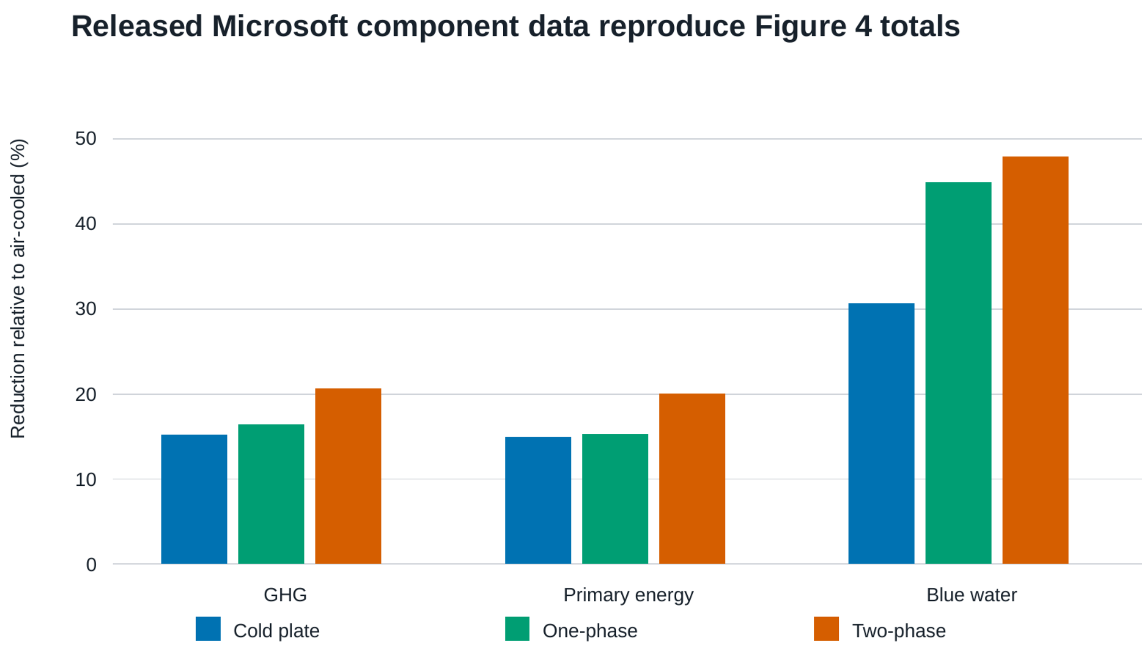

Figure 2. Reconstructed reductions relative to the air-cooled case for the grid scenario in the released Microsoft/Nature source data. Values are derived from published normalized component contributions and validate arithmetic consistency, not proprietary background inventories or foreground bills of materials.

For primary energy, the released data gave reductions of 14.98%, 15.33%, and 20.05% for cold plate, one-phase immersion, and two-phase immersion, respectively. Corresponding GHG reductions were 15.18%, 16.41%, and 20.66%. Blue-water reductions were larger: 30.64%, 44.94%, and 47.88%. Under the released 100% renewable scenario, the normalized GHG results were substantially lower for all architectures, while the relative ranking and contribution mix remained dependent on embodied and non-electricity terms.

4.2. Grid intensity changes the cold-plate/one-phase ranking

Figure 3 applies Eq. (7) to the 51 eGRID state/DC records. The modeled cold-plate and one-phase totals intersect at 60.8 kg CO_2e MWh^{-1}, within the observed 23.7-893.1 kg CO_2e MWh^{-1} state range. Cold plate was lower than one-phase immersion only for Vermont (23.7 kg CO_2e MWh^{-1}); one-phase was lower for the other 50 records. At the Arkansas factor (452.9 kg CO_2e MWh^{-1}), one-phase was lower than cold plate by 0.617 kg CO_2e $Vcore^{-1}$ yr^{-1}. Two-phase immersion remained the lowest-GHG architecture in all 51 re-based cases under the released assumptions.

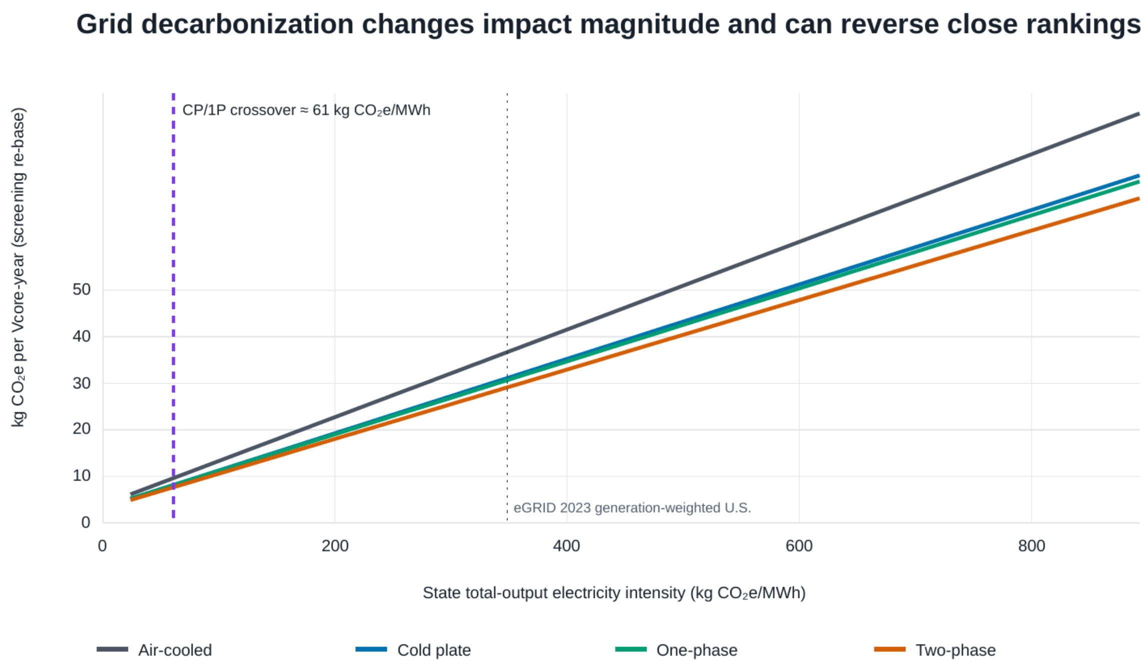

Figure 3. Screening re-basing of the released Microsoft/WSP GHG contributions across EPA eGRID 2023 state total-output intensities. The vertical dashed line marks the cold-plate/one-phase crossover near 61 kg CO2e MWh−1. Lines connect state scenarios and do not represent a dynamic grid trajectory. Two-phase immersion remains lowest under the released foreground assumptions.

The result has two implications. First, the statement that one liquid-cooling architecture is categorically preferable does not follow from a national- average case: at least one pair changes order within contemporary U.S. grids. Second, grid intensity is not the only determinant. Two-phase immersion remained lowest because the released model combined lower use-phase burden with Vcore-normalized server and building effects. This ranking should not be transferred to fluids, server designs or regulations outside the released foreground assumptions.

4.3. Decarbonization transfers attention from operation to hardware

At the high end of the eGRID state range, embodied contributions were approximately 4.0-4.2% of total GHG across architectures. At the lowest state factor they increased to 56.9-58.2%. Thus, grid decarbonization does not make cooling architecture irrelevant; it changes the evidence required to judge it. PUE and IT power dominate high-carbon locations, whereas server quantity, server lifetime, overclocking, electronics manufacturing and replacement become co-dominant on low-carbon grids.

The released endpoints illustrate the same transition. In the grid case, use phase contributed approximately 90% of mean GHG. In the renewable case, the absolute use-phase term fell sharply while the embodied server terms were unchanged. An operational-only comparison can therefore be directionally useful on a carbon-intensive grid but becomes increasingly incomplete as electricity decarbonizes.

4.4. Contribution-weighted pedigree scores identify the next evidence

The released pedigree assessment alone assigns the weakest average score to networking equipment (3.6/5), followed by storage servers (2.8/5). Weakness alone, however, would prioritize a small contributor. Multiplying weakness by mean GHG contribution changed the order (Figure 4). Use phase ranked first because its 90.3% mean contribution outweighed a relatively good 1.4/5 pedigree score. Storage, networking and compute-server evidence formed the next tier with nearly equal priority indices (0.0127-0.0129).

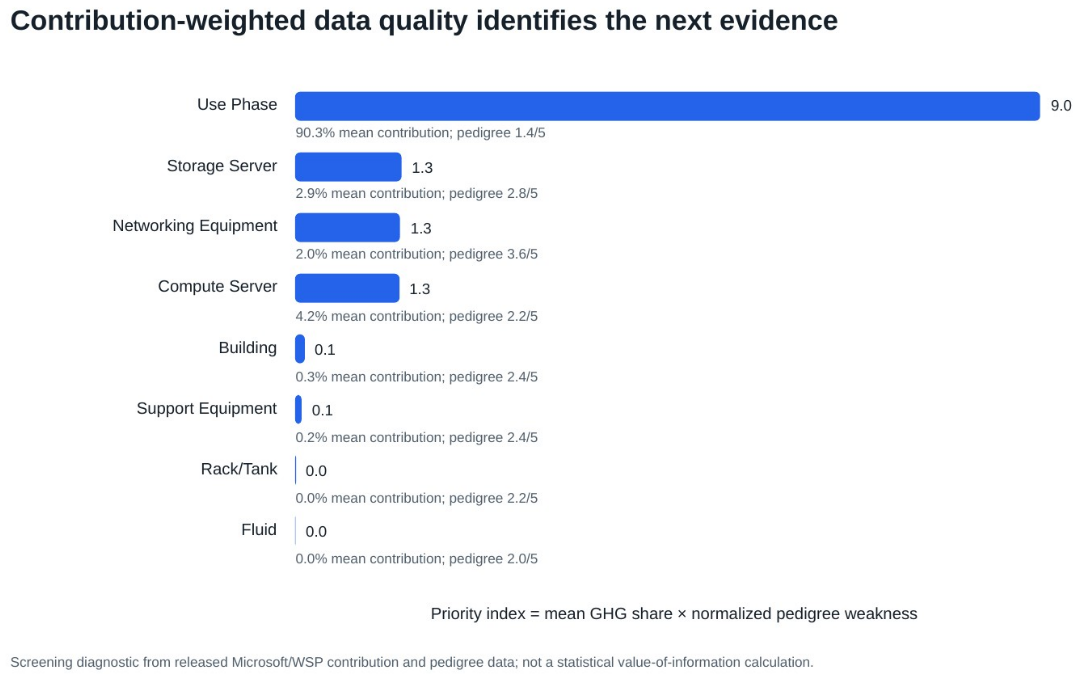

Figure 4. Contribution-weighted evidence-improvement priority derived from the released Microsoft/WSP grid-case GHG contributions and pedigree matrix. The diagnostic multiplies mean contribution share by normalized pedigree weakness; it is a transparent screening rank, not a formal expected value of information.

This result separates two research programs. Improving measured PUE, IT power, utilization and water response has the greatest present consequence for grid-powered facilities. Improving server inventories—especially storage and networking proxies—becomes the next priority and grows in importance on cleaner grids. Fluid data remain important for toxicity, regulation, leakage and feasibility even though their released climate contribution is small; the priority index is not a complete environmental, health and safety ranking.

4.5. Public product and construction data quantify uncertainty and leverage

The 48 usable Boavizta server records yielded manufacturing GHG values from 465 to 2,503 kg CO2e per server, with a median of 1,215 kg CO2e and an interquartile range of 1,146-1,337 kg CO2e (Figure 5). Annualized by the declared four-year lifetime, the median was 304 kg CO2e server^{-1} yr^{-1}. The approximately 5.4-fold full range is large relative to many reported cooling savings. Because manufacturer PCFs differ in method and configuration, this range is an evidence envelope rather than a product ranking.

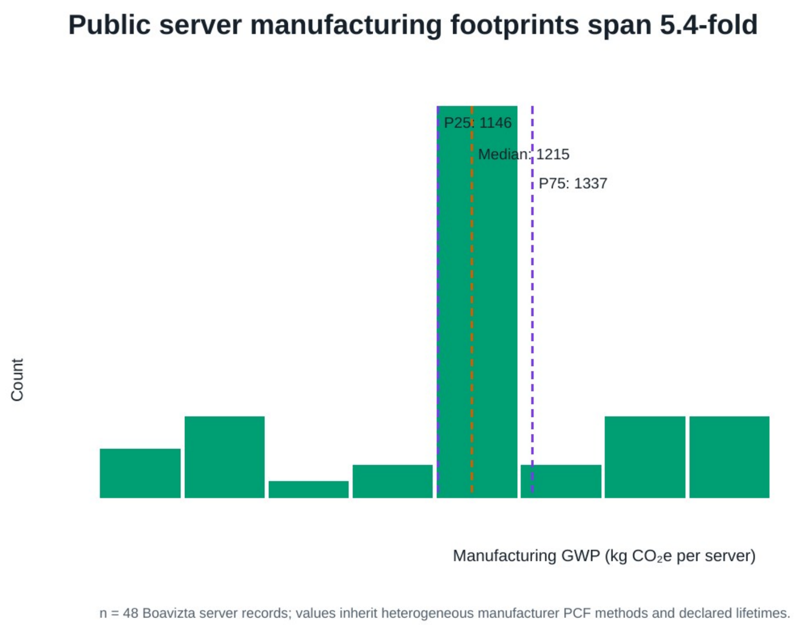

Figure 5. Distribution of manufacturing GHG for 48 Boavizta `Datacenter/Server` records having total product GHG and manufacturing share. Dashed lines show the 25th percentile, median and 75th percentile. Heterogeneous manufacturer methods and product configurations preclude a product ranking.

The selected ÖKOBAUDAT records identified large construction procurement levers (Figure 6). High-scrap electric-arc-furnace galvanized steel had 59.9% lower A1-A3 GHG, 47.9% lower nonrenewable primary energy and 52.0% lower freshwater use than the selected low-scrap blast-furnace profile. CEM III cement had 49.8% lower GHG, 12.1% lower nonrenewable primary energy and 23.5% lower freshwater use than CEM II/A. Their facility significance cannot be calculated without architecture-specific material quantities.

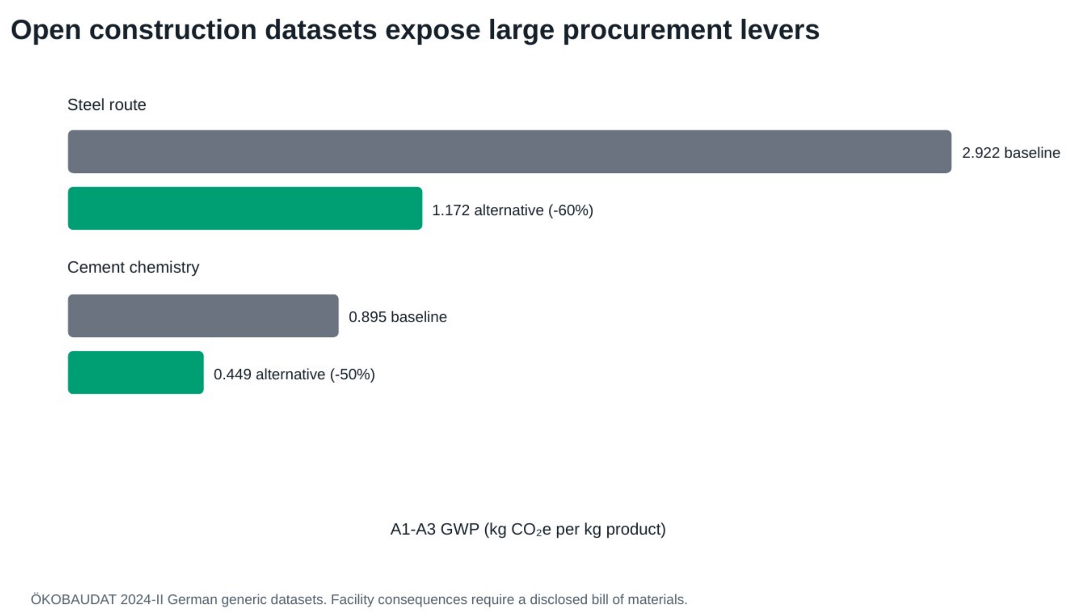

Figure 6. Selected ÖKOBAUDAT 2024-II A1-A3 GHG factors for baseline and alternative steel and cement routes. The figure identifies unit-process procurement leverage; it does not quantify a data-center reduction because foreground quantities are unavailable.

4.6. Temporal and measurement pathways remain validation cases

The Fayetteville TMY and assumed temperature-PUE curves produced annual mean PUE values of 1.0907, 1.0529, 1.0406 and 1.0352 for air, cold plate, one-phase and two-phase cases, respectively. These are hypotheses, not technology findings. Likewise, the synthetic measurement surfaces processed 8,760 hourly observations, rejected extrapolation and propagated declared point uncertainty. Their role is to specify exactly how reviewed laboratory data will enter the lifecycle calculation. The resulting climate and surface figures are retained as Supplementary Figures rather than core evidence.

4.7. Capability and evidence matrix

Table 1 summarizes the implemented functions and the evidence required for their defensible use. The distinguishing contribution is the continuity from data registration through physics-informed temporal integration to an explicit claim gate. Numerical outputs remain available for inspection even when the audit blocks a comparative conclusion.

Table 1. OpenDC-LCA capability and evidence matrix.

Capability	Input	Method	Output	Evidence requirement	Status
Source registration	Provider, ID, version, geography, unit, boundary, license	Schema validation and checksum manifest	Traceable record	Complete metadata and redistribution review	Implemented
Static lifecycle screening	Energy, PUE, grid factors, equipment, fluids, water	Annual energy balance and lifecycle annualization	Absolute and normalized impacts	Compatible unit and boundary	Implemented
Temporal operation	Hourly weather, workload, grid and cooling response	Hourly alignment and integration	PUE, energy, water, operational GHG	Full coverage and accounting basis	Implemented

Capability	Input	Method	Output	Evidence requirement	Status
Performance surface	Load-temperature grid, uncertainty, source IDs	Bounded bilinear interpolation	Hourly cooling response	Reviewed measurements; no extrapolation	Implemented
Inventory exchange	openLCA JSON-LD processes or OpenDC-LCA scenario	Exchange-preserving import/export and Brightway mapping	Foreground process network	Flow, provider, unit, system-model and LCIA compatibility review	Implemented
Reliability	Weibull/Arrhenius parameters, maintenance and downtime	Renewal equation and lifecycle annualization	Expected failures, replacements, downtime and impacts	Empirical failure model and uncertainty	Implemented
Uncertainty and sensitivity	Distributions, seed and perturbations	Monte Carlo and elasticity	Intervals and ranked drivers	Distribution and correlation review	Screening
Scientific claim gate	Scenario, sources, evidence status and intent	Automated blockers and warnings	Reviewable findings	Independent critical review for public claims	Implemented
Access	JSON or CSV	Shared engine through CLI, API and local GUI	JSON, tables, figures and reports	Same model version across interfaces	Implemented

4.8. Practitioner workflow and release verification

The v1.1.0 practitioner path reduces the minimum viable screening study to the input-output contract in Table 2. It is available through four command-line operations (`capabilities`, `new-study`, `prepare-study`, and `practitioner-report`), corresponding Python API calls, and the guided local interface. Both practitioner interfaces first create the full governed scenario and then call the same analysis and audit functions used by the advanced workflow. Thus, ease of entry does not bypass provenance, unit, boundary, or evidence checks.

Table 2. Compact practitioner input-output contract in OpenDC-LCA v1.1.0.

Stage	User-supplied information	Machine action	Principal output
Define facility	Study name, cooling architecture, geography, reference year, IT capacity, capacity factor, PUE, lifetime and onsite water	Validate physical ranges and compute annual IT and facility electricity	Transparent operating basis
Add electricity	Location-based GHG, primary-energy and blue-water factors with compatible accounting basis	Apply factors to annual facility electricity	Annual and per-IT-MWh operational impacts
Add equipment, if known	Quantity, service life, production and end-of-life impacts	Annualize production and replacement over the study period	Operational-plus-equipment totals and contribution shares
Register evidence	Source ID, title, citation, license, quality, uncertainty, review and confidentiality status	Preserve metadata in the governed scenario and run the scientific audit	Traceable evidence level, warnings and blockers
Export decision record	No additional input	Serialize the same calculation and audit result for people and software	`PRACTITIONER_REPORT.md`, `practitioner-results.json`, model version and scenario SHA-256 digest

The public release was checked at three levels. First, 40 automated tests cover the calculation engine, provenance and claim gate, performance maps, uncertainty, reports, interoperability, reliability, API, GUI endpoints, and the end-to-end practitioner commands. Second, the wheel was installed in a clean Python 3.12 environment and the practitioner workflow was executed from template creation through report generation. Third, the tagged GitHub release contains both wheel and source-distribution artifacts produced by the release workflow. These checks establish installation and software consistency; they do not constitute ISO critical review or validate user-supplied factors.

5. Discussion

5.1. The technology ranking is conditional, not universal

The geographic screening exposes a decision boundary that is hidden by a single national-average scenario. Cold plate and one-phase immersion cross at 60.8 kg CO2e MWh−1: a small difference in their non-use-phase terms reverses the ordering once grid emissions are sufficiently low. The practical take-home message is not that Vermont should select cold plates. The affine transfer holds the released foreground model fixed and changes only the electricity-dependent term. Rather, the finding demonstrates that a cooling ranking must be reported together with its grid range, foreground assumptions and crossover conditions. The same principle applies to climate, workload, water stress and equipment lifetime.

Two-phase immersion did not cross another architecture in the tested state range. That result is stronger within the released model but is not universal. It inherits the paper's assumptions concerning server count and performance, fluid production and loss, cooling-system boundaries, equipment lifetime and regulatory feasibility. A robust comparative claim should therefore report both the observed no-crossover range and the untested assumptions that could create one.

5.2. Decarbonization changes the research question

At the highest observed state grid intensity, embodied terms supplied only about 4% of the re-based total; at the lowest, they supplied 57-58%. This transition changes what constitutes adequate evidence. On a carbon-intensive grid, improved PUE, reduced IT power and water-aware heat rejection are the largest near-term levers. On a low-carbon grid, the comparison becomes increasingly sensitive to useful computation per server, server count, manufacturing, lifetime and replacement. Operational efficiency and embodied impact are therefore not competing narratives but successive constraints along a decarbonization pathway.

The 5.4-fold range in public Boavizta server manufacturing footprints shows why a single generic server factor is inadequate for low-carbon scenarios. It does not establish a probability distribution or rank products, because the records are heterogeneous. It does establish scale: public foreground dispersion can exceed the differences among cooling options. Reporting a cooling saving to high precision while proxying the servers with one legacy factor creates false confidence. Product-specific bills of materials, functional performance and harmonized product-carbon-footprint rules are thus central research data, not secondary documentation.

5.3. Evidence priority links LCA to experiments and procurement

The contribution-weighted pedigree analysis provides an actionable measurement sequence. Present grid-powered comparisons should first improve hourly IT and cooling electricity, utilization, part-load parasitics and on-site water response. Storage, networking and compute-server inventories form the next evidence tier. As grids decarbonize, this second tier moves upward. This ranking is intentionally transparent and can be replaced by formal value-of- information analysis when parameter distributions and decision losses become available.

For electronics-cooling laboratories, the immediate contribution is a shared performance-map protocol spanning heat load, ambient and coolant temperature, flow, pressure drop, pump and fan power, heat-rejection power, water, duration, uncertainty and calibration lineage. A complete bounded surface permits annual integration without extrapolation. Comparative experiments should use a common heat-load emulator and heat-rejection boundary and should separate IT-integral fan power from facility parasitics. Those measurements would replace the present synthetic fixtures through the same data contract.

The ÖKOBAUDAT comparison identifies a complementary procurement pathway. Selected lower-impact steel and cement routes reduced cradle-to-gate GHG by approximately 50-60% per kilogram. Whether this matters more than cooling equipment or server replacement depends on an architecture-specific bill of quantities, which is currently absent. The appropriate next step is therefore not to apply the percentages to an assumed facility, but to obtain reviewed quantities and preserve product geography, unit and module boundary.

5.4. Role relative to openLCA and Brightway

OpenDC-LCA is not a replacement for openLCA or Brightway [3,32]. Those platforms manage process networks, databases and impact assessment. The contribution here is a data-center-specific foreground and evidence layer: cooling-performance surfaces, hourly operation, reliability and replacement, functional-unit controls, source lineage and a comparative-claim gate. The interoperability functions preserve identifiers and expose missing providers or impact methods rather than silently treating an imported archive as a complete product system.

This separation also explains why the downloaded GLAD hydrogen inventories and USLCI database were not forced into the reported totals. They are relevant to backup power and equipment supply chains, but numerical combination without provider linking, a shared system model and a selected LCIA method would create an apparently comprehensive but irreproducible result. Evidence registration, successful exchange and impact calculation are three distinct stages.

5.5. Limitations and decision-grade next steps

The state analysis is a screening re-basis between two released endpoints, not a 51-state process LCA. It uses annual location-based eGRID factors and omits hourly or marginal emissions, procurement contracts and transmission effects. The exact reconstruction verifies released arithmetic, not licensed ecoinvent or LCA for Experts inventories, confidential bills of materials or measured PUE. The Boavizta sample mixes manufacturers and methods; the ÖKOBAUDAT factors are German product-stage proxies without data-center quantities. The USGS file contains watershed geometry rather than withdrawal, consumption or scarcity characterization, so no watershed-level water claim is made.

The temporal examples use TMY weather and declared hypothetical or synthetic performance curves. They exclude humidity-sensitive heat rejection, extreme events and correlated measurement or model-form error. Reliability results are expectations from Weibull renewal and optional Arrhenius acceleration; they exclude redundancy, dependent failures and repair queues. Vcore-year improves on rack- or facility-level comparisons but does not fully represent useful computation, workload quality or rebound effects. Automated checks cannot replace an ISO-aligned critical review.

Four additions would enable a decision-grade multi-location comparison: (i) reviewed architecture-specific performance surfaces and water measurements; (ii) server, network, storage, facility and cooling-system bills of quantities with uncertainty and lifetime data; (iii) hourly electricity and watershed scarcity characterization; and (iv) harmonized background product systems in openLCA or Brightway. Dr. Nutter's independent review should assess the recorded methodological decisions, dataset mappings and claim-gate findings as a coherent package before public comparative assertions are made.

5.6. Practitioner use and interpretation boundary

The guided workflow makes the framework usable before a complete bill of materials is available, but it labels the consequence. A study without equipment data is identified as operational-only; a zero factor is treated as zero rather than as an undocumented unknown; and omitted fluids, reliability, hourly variation, scarcity characterization, or compute-performance effects remain visible in the capability boundary and next-data list. This is a deliberate adoption strategy: practitioners can obtain a reproducible screening result immediately while receiving an explicit acquisition plan for decision-grade evidence.

The human-readable and JSON outputs serve different users without creating two scientific records. The report supports engineering review and procurement discussion, whereas the JSON result supports portfolio aggregation, dashboards, scripted comparisons, and archival. Model version and scenario digest make the result attributable to a specific calculation and governed input. Comparative communication remains conditional on the automated claim gate and, for public assertions, independent critical review.

6. Conclusions

OpenDC-LCA converts heterogeneous cooling, grid, server, construction and data- quality evidence into conditional, auditable lifecycle comparisons. It exactly reconstructed 24 released Microsoft/Nature totals and then exposed a result that a single reference scenario cannot show: cold plate and one-phase immersion cross near 60.8 kg CO_2e MWh^{-1} within the observed U.S. state range. Across that range, the embodied share rose from about 4% to 57-58%, shifting the limiting evidence from operational electricity toward servers and replacement. Forty-eight public server records spanned 465-2,503 kg CO_2e of manufacturing emissions, while selected lower-impact steel and cement routes showed approximately 50-60% cradle-to-gate GHG leverage per kilogram.

These findings support three conclusions. First, cooling rankings should be reported as decision surfaces with crossover conditions, not universal league tables. Second, decarbonized operation makes server performance, manufacturing, lifetime and architecture-specific material quantities first-order LCA data. Third, contribution-weighted data quality can convert an LCA into a prioritized experimental and data-acquisition program. OpenDC-LCA operationalizes these principles through traceable source registration, temporal performance maps, reliability, openLCA/Brightway exchange and machine-readable claim gates. The released v1.1.0 package also provides a guided path from a compact practitioner input to a governed scenario, plain-language report, machine-readable result, evidence classification, and next-data plan. This makes the framework usable for screening while preserving a visible boundary between calculated, omitted, and insufficiently supported effects. The remaining barrier to a public decision-grade benchmark is evidence: harmonized bills of quantities, measured operating surfaces, time- and watershed-resolved factors, and independent critical review.

Data and code availability

OpenDC-LCA version 1.1.0 source code, examples, documentation, tests, and derived paper tables and figures are available at https://github.com/UARK-NED3/OpenDC-LCA. Installable wheel and source archives are preserved in the tagged release at https://github.com/UARK-NED3/OpenDC-LCA/releases/tag/v1.1.0. The repository documents the guided interface, compact input contract, advanced schema, command-line interface, Python API, report fields, and current capability boundary. Provider-native raw files remain local when redistribution permission has not been established. Released Microsoft/Nature model files are available from Zenodo [2]. EPA eGRID and NOAA TMY data are available from their respective public portals [5,6].

Author contributions

Draft taxonomy for confirmation: Han Hu: conceptualization, methodology, software supervision, thermal-management domain analysis, writing, and visualization. Darin W. Nutter: LCA methodology review, validation, critical review, and manuscript revision. Contributions must be confirmed before submission.

Competing interests

The authors must complete and approve the journal-specific competing-interest statement before submission.

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